

Topic 02: Definition of Laplace Transform

Unit IV — Laplace Transforms

JNTUA College of Engineering (Autonomous) ATP

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1 Opening Hook

Hook

Every powerful machine needs a precise blueprint. The Laplace Transform is no exception. Until now, we know *why* it exists — to convert calculus into algebra. But *what exactly* is it? It turns out to be a single integral with a very clever choice of kernel: e^{-st} . This one integral, evaluated from 0 to infinity, encodes everything about $f(t)$ into a new function $F(s)$. Let's take the blueprint apart, symbol by symbol.

2 Why This Topic Exists

Before we begin, here is the honest answer to why you are reading this...

Without the formal definition, you are using the Laplace Transform as a black box — you can apply it but not derive new results or extend it. The table below shows what happens when students skip this foundation.

Mistake	Consequence
Skipping the definition	Unable to compute transforms of non-standard functions
Confusing $f(t)$ with $F(s)$ notation	Writing wrong answers in every subsequent topic
Not understanding the convergence parameter s	Missing why s has a lower bound
Treating the integral as a finite definite integral	Arriving at incorrect limits of integration
Forgetting that s is a parameter, not a variable like t	Algebraic errors when collecting s -terms

Why Foundations Matter

Every formula you will use in Units IV and V descends directly from one integral. If that integral is clear to you, the rest of the course flows naturally. Skip it and you will be memorising without understanding.

3 You Already Know This (Intuition First)

Imagine a weighing machine. You place an object — call it $f(t)$ — on the scale. The scale is the integral $\int_0^\infty e^{-st}(\cdot) dt$. The reading on the dial is $F(s)$.

Now here is the key insight: changing the value of s is like switching the units on the dial from grams to kilograms. You get a different number for each s , but the *object* $f(t)$ has not changed at all. The kernel e^{-st} is the “calibration knob” that ensures the reading is finite (convergent) for suitably large s .

This is why $F(s)$ is called a **transform**: it is the same information as $f(t)$, re-expressed in a new language. The two domains are genuinely different — t is time (measured in seconds), while s has units of 1/second, placing it in the **frequency domain**.

4 The Formal Definition

Definition of the Laplace Transform

Let $f(t)$ be a function defined for $t \geq 0$. The **Laplace Transform** of $f(t)$ is the function $F(s)$ defined by

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt,$$

provided the improper integral converges.

Symbol-by-symbol unpacking:

- $f(t)$: the original function, defined for $t \geq 0$ (time domain).
- e^{-st} : the *damping kernel* — it forces the integrand to decay to zero as $t \rightarrow \infty$, ensuring convergence for suitably large s .
- $\int_0^{\infty}$: an improper integral — formally the limit $\lim_{T \rightarrow \infty} \int_0^T e^{-st} f(t) dt$.
- $F(s)$: the Laplace transform — a brand-new function of the variable s (s-domain).
- s : a complex parameter; for most B.Tech applications treat it as a real positive number.

Because the transform is built from an integral, it is called an **integral transform**. In engineering, the s -domain is also called the **frequency domain**.

What Did We Learn?

The Laplace Transform converts a time-domain function $f(t)$ into an s -domain function $F(s)$ via a single improper integral with exponential kernel e^{-st} . The parameter s must be large enough to guarantee convergence.

5 Notation Conventions

Strict Notation Rules Used Throughout This Course

- Time-domain functions use **lowercase**: $f(t)$, $g(t)$, $y(t)$.
- Transform-domain functions use **uppercase**: $F(s)$, $G(s)$, $Y(s)$.
- The transform operator is always written as $\mathcal{L}\{f(t)\}$ — use the calligraphic $\mathcal{L}$, *never* the plain letter L .

- Transform pair notation: $f(t) \xleftrightarrow{\mathcal{L}} F(s)$.

Common notation errors and their corrections:

Incorrect Notation	Correct Notation
$L\{f(t)\}$	$\mathcal{L}\{f(t)\}$
$\mathcal{L}\{F(t)\}$	$\mathcal{L}\{f(t)\}$ (use lowercase for time domain)
$f(s)$ as the transform result	$F(s)$ (uppercase for s -domain)
$F(t)$ for the transform	$F(s)$ (the transform is a function of s , not t)
$\mathcal{L}(f(t))$ (round brackets)	$\mathcal{L}\{f(t)\}$ (curly braces)
$\mathcal{L}^{-1}(F(s))$	$\mathcal{L}^{-1}\{F(s)\}$ (curly braces here too)
$L^{-1}\{F(s)\}$	$\mathcal{L}^{-1}\{F(s)\}$ (calligraphic always)

What Did We Learn?

Notation is not cosmetic. Mixing uppercase and lowercase, or using plain L instead of $\mathcal{L}$, leads to wrong answers that are impossible to track down later. Commit these conventions now.

6 Linearity of the Laplace Transform

Linearity Property — Statement and Proof

Theorem (Linearity). Let $f(t)$ and $g(t)$ be functions whose Laplace transforms exist, and let a, b be any constants. Then

$$\mathcal{L}\{a f(t) + b g(t)\} = a \mathcal{L}\{f(t)\} + b \mathcal{L}\{g(t)\} = a F(s) + b G(s).$$

Proof (directly from the definition):

Starting from the definition,

$$\begin{aligned} \mathcal{L}\{a f(t) + b g(t)\} &= \int_0^{\infty} e^{-st} [a f(t) + b g(t)] dt \\ &= \int_0^{\infty} [a e^{-st} f(t) + b e^{-st} g(t)] dt && \text{(distribute } e^{-st}) \\ &= a \int_0^{\infty} e^{-st} f(t) dt + b \int_0^{\infty} e^{-st} g(t) dt && \text{(linearity of the integral)} \\ &= a F(s) + b G(s). && \text{(definition applied twice)} \end{aligned}$$

□

Why linearity matters. Any linear combination of functions — such as $3e^{2t} - 5 \cos t + 7$ — can be split into individual pieces, each transformed separately, then the results added. This is the engine behind every table of transforms you will use in this course.

What Did We Learn?

Linearity means “break, transform, and add.” It is the reason a short table of basic transforms is enough to handle enormously complicated functions.

7 First Direct Computations

Derivation 1 — $\mathcal{L}\{1\}$

We compute directly from the definition:

$$\begin{aligned} \mathcal{L}\{1\} &= \int_0^{\infty} e^{-st} \cdot 1 dt \\ &= \lim_{T \rightarrow \infty} \int_0^T e^{-st} dt \\ &= \lim_{T \rightarrow \infty} \left[\frac{e^{-st}}{-s} \right]_0^T \\ &= \lim_{T \rightarrow \infty} \left(\frac{e^{-sT}}{-s} - \frac{e^0}{-s} \right) \\ &= \lim_{T \rightarrow \infty} \left(-\frac{e^{-sT}}{s} + \frac{1}{s} \right). \end{aligned}$$

Since $s > 0$, we have $e^{-sT} \rightarrow 0$ as $T \rightarrow \infty$. Therefore

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad s > 0.$$

What Did We Learn?

The constant function 1 transforms to $1/s$. The condition $s > 0$ is necessary: if $s \leq 0$ the exponential e^{-sT} would not decay to zero and the integral would diverge.

Derivation 2 — $\mathcal{L}\{e^{at}\}$

$$\begin{aligned} \mathcal{L}\{e^{at}\} &= \int_0^\infty e^{-st} e^{at} dt \\ &= \int_0^\infty e^{-(s-a)t} dt \\ &= \lim_{T \rightarrow \infty} \int_0^T e^{-(s-a)t} dt \\ &= \lim_{T \rightarrow \infty} \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^T \\ &= \lim_{T \rightarrow \infty} \left(-\frac{e^{-(s-a)T}}{s-a} + \frac{1}{s-a} \right). \end{aligned}$$

For the limit to be zero we need $s - a > 0$, i.e. $s > a$. Under that condition $e^{-(s-a)T} \rightarrow 0$, giving

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad s > a.$$

Why $s > a$? If $s \leq a$ then $e^{-(s-a)t} = e^{|s-a|t}$ grows without bound and the integral diverges. The condition $s > a$ is the *convergence strip* for this transform.

What Did We Learn?

$\mathcal{L}\{e^{at}\} = 1/(s-a)$ for $s > a$. The sign in the denominator is $s-a$ (minus), *not* $s+a$. This is the single most common sign error in this course.

Derivation 3 — $\mathcal{L}\{t\}$

$$\mathcal{L}\{t\} = \int_0^\infty e^{-st} t dt.$$

We apply integration by parts. Let $u = t$ and $dv = e^{-st} dt$. Then $du = dt$ and

$v = -\frac{1}{s}e^{-st}$. Therefore

$$\begin{aligned}\int_0^\infty t e^{-st} dt &= \left[t \cdot \left(-\frac{e^{-st}}{s} \right) \right]_0^\infty - \int_0^\infty \left(-\frac{e^{-st}}{s} \right) dt \\ &= \left[-\frac{t e^{-st}}{s} \right]_0^\infty + \frac{1}{s} \int_0^\infty e^{-st} dt.\end{aligned}$$

Evaluate the boundary term: as $t \rightarrow \infty$ with $s > 0$, $t e^{-st} \rightarrow 0$ (exponential dominates the linear factor); at $t = 0$, $t e^{-st} = 0$. So the boundary term is $0 - 0 = 0$. For the remaining integral we already know $\int_0^\infty e^{-st} dt = 1/s$. Hence

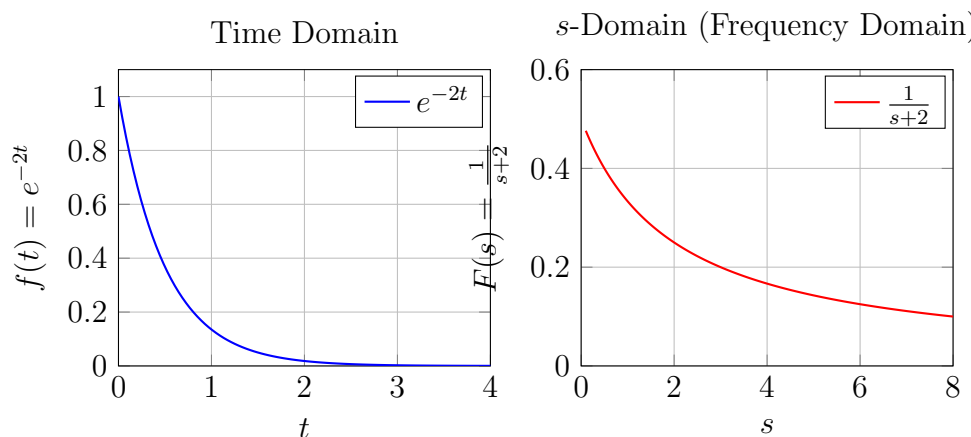
$$\mathcal{L}\{t\} = 0 + \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2}, \quad s > 0.$$

What Did We Learn?

Integration by parts is the standard technique for transforms of polynomial terms. The boundary term vanishes because the exponential e^{-st} decays faster than any power of t grows — a fact that will recur throughout this course.

8 Visualizing the Transform

The two plots below illustrate the same “information” expressed in two completely different domains. The left plot shows $f(t) = e^{-2t}$ in the time domain; the right shows its Laplace transform $F(s) = \frac{1}{s+2}$ in the s -domain.



Caption: The same “information” lives in both plots — but $F(s)$ is not a plot of $f(t)$ vs. s . They are completely different functions in completely different domains.

What Did We Learn?

The time-domain plot shows exponential decay. The s -domain plot shows a hyperbola decreasing toward zero. Both encode the same physical signal; the Laplace Transform simply re-expresses it in a form that makes algebra easier.

9 Worked Examples

Example 1 — $\mathcal{L}\{5 - 3e^{2t}\}$

Objective: Use linearity and the results of Section 7 to compute this transform without re-deriving from first principles.

Step 1. Apply linearity:

$$\mathcal{L}\{5 - 3e^{2t}\} = \mathcal{L}\{5 \cdot 1\} + \mathcal{L}\{(-3)e^{2t}\} = 5\mathcal{L}\{1\} - 3\mathcal{L}\{e^{2t}\}.$$

Step 2. Substitute the known results ($a = 0$ for the constant, $a = 2$ for the exponential):

$$= 5 \cdot \frac{1}{s} - 3 \cdot \frac{1}{s - 2}.$$

Step 3. Combine over a common denominator:

$$= \frac{5(s - 2) - 3s}{s(s - 2)} = \frac{5s - 10 - 3s}{s(s - 2)} = \frac{2s - 10}{s(s - 2)}.$$

Convergence condition: The transform of e^{2t} requires $s > 2$, which is the binding constraint. The transform of the constant 1 requires only $s > 0$. Therefore the combined result is valid for $s > 2$.

$$\boxed{\mathcal{L}\{5 - 3e^{2t}\} = \frac{2s - 10}{s(s - 2)}, \quad s > 2.}$$

What Did We Learn?

When computing the transform of a linear combination, always: (1) split by linearity, (2) apply known results, (3) state the convergence condition as the *most restrictive* condition among the individual terms.

Example 2 — $\mathcal{L}\{e^{-t} + 4\}$

Step 1. Apply linearity:

$$\mathcal{L}\{e^{-t} + 4\} = \mathcal{L}\{e^{-t}\} + 4\mathcal{L}\{1\}.$$

Step 2. Use $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$ with $a = -1$:

$$\mathcal{L}\{e^{-t}\} = \frac{1}{s - (-1)} = \frac{1}{s + 1}, \quad s > -1.$$

And $\mathcal{L}\{1\} = \frac{1}{s}$ for $s > 0$.

Step 3. Combine:

$$\mathcal{L}\{e^{-t} + 4\} = \frac{1}{s + 1} + \frac{4}{s} = \frac{s + 4(s + 1)}{s(s + 1)} = \frac{5s + 4}{s(s + 1)}.$$

Convergence condition: $s > 0$ is the binding constraint (more restrictive than $s > -1$).

$$\mathcal{L}\{e^{-t} + 4\} = \frac{5s + 4}{s(s + 1)}, \quad s > 0.$$

What Did We Learn?

When a is negative (here $a = -1$), the formula $1/(s - a)$ gives $1/(s + 1)$ — the denominator becomes s *plus* a positive number. Always substitute the sign of a explicitly rather than memorising by pattern.

Example 3 — Identifying a Sign Error

Problem. A student writes

$$\mathcal{L}\{e^{at}\} = \frac{1}{s + a}.$$

Identify the error and give the correct answer.

Error identification. The student has written $s + a$ in the denominator. The correct derivation (see Section 7, Derivation 2) shows that the combined exponent in the integral is $-(s - a)t$, giving denominator $(s - a)$. Writing $s + a$ is equivalent to claiming the exponent is $-(s + a)t$, which corresponds to the transform of e^{-at} , not e^{at} .

Where the error likely comes from. Students remember the exponential “goes to the bottom” and place the letter a after s without tracking the sign. The safest method: always merge the two exponentials first: $e^{-st} \cdot e^{at} = e^{-(s-a)t}$, and then integrate directly.

Correct answer:

$$\mathcal{L}\{e^{at}\} = \frac{1}{s - a}, \quad s > a.$$

What Did We Learn?

The denominator for $\mathcal{L}\{e^{at}\}$ is always $s - a$ (minus), never $s + a$. A quick sanity check: at $a = 0$ the formula should reduce to $1/(s - 0) = 1/s = \mathcal{L}\{1\}$, which is correct.

10 Excel Example

Objective: Numerically verify that $\mathcal{L}\{e^{-2t}\}$ at $s = 5$ equals $1/(5 + 2) = 1/7 \approx 0.1429$.

The integral to approximate is

$$\int_0^\infty e^{-5t} \cdot e^{-2t} dt = \int_0^\infty e^{-7t} dt.$$

We discretise from $t = 0$ to $t = 5$ in steps of $h = 0.5$ and apply the **trapezoid rule**:

$$\text{Area} \approx \frac{h}{2} [f(t_0) + 2f(t_1) + \cdots + 2f(t_{n-1}) + f(t_n)].$$

Column A	Column B	Column C
t	$e^{-5t} \cdot e^{-2t}$	Cumulative Trapezoid Area
0	=EXP(-5*A2)*EXP(-2*A2)	(start at 0)
0.5	=EXP(-5*A3)*EXP(-2*A3)	=C2+(A3-A2)/2*(B2+B3)
1.0	=EXP(-5*A4)*EXP(-2*A4)	=C3+(A4-A3)/2*(B3+B4)
1.5	=EXP(-5*A5)*EXP(-2*A5)	=C4+(A5-A4)/2*(B4+B5)
2.0	=EXP(-5*A6)*EXP(-2*A6)	=C5+(A6-A5)/2*(B5+B6)
2.5	=EXP(-5*A7)*EXP(-2*A7)	=C6+(A7-A6)/2*(B6+B7)
3.0	=EXP(-5*A8)*EXP(-2*A8)	=C7+(A8-A7)/2*(B7+B8)
3.5	=EXP(-5*A9)*EXP(-2*A9)	=C8+(A9-A8)/2*(B8+B9)
4.0	=EXP(-5*A10)*EXP(-2*A10)	=C9+(A10-A9)/2*(B9+B10)
4.5	=EXP(-5*A11)*EXP(-2*A11)	=C10+(A11-A10)/2*(B10+B11)
5.0	=EXP(-5*A12)*EXP(-2*A12)	=C11+(A12-A11)/2*(B11+B12)

How to interpret the spreadsheet: Column B evaluates e^{-7t} at each row (note: $\text{EXP}(-5*A2)*\text{EXP}(-2*A2) = e^{-7 \times 0} = 1$ at $t = 0$, and approaches 0 rapidly). Column C accumulates the trapezoid area strip by strip. Because e^{-7t} decays very rapidly, the first strip alone ($t = 0$ to 0.5) yields an area of $\frac{0.5}{2}(1 + e^{-3.5}) \approx 0.2575$. The cumulative sum with step size $h = 0.5$ reaches **0.2656** at $t = 5$ (an overestimation of $\approx 86\%$ above the exact value $1/7 \approx 0.1429$ because a linear chord overestimates a strictly convex decaying curve over a coarse interval). To approach **0.1429** accurately, the step size in Excel must be refined (e.g., $h \leq 0.05$, where $h = 0.05$ yields 0.1443 and $h = 0.01$ yields 0.1429).

What Did We Learn?

The exact analytical value is $\mathcal{L}\{e^{-2t}\}|_{s=5} = \frac{1}{5+2} = \frac{1}{7} \approx 0.1429$. For rapidly decaying functions like e^{-7t} , coarse discretisation ($h = 0.5$) gives 0.2656; refining the step size to $h \leq 0.05$ brings the numerical trapezoid sum to 0.1429. The tail beyond $t = 5$ contributes less than $e^{-35} \approx 6 \times 10^{-16}$ and is negligible.

11 Python Example

The script below uses **sympy** to compute the Laplace Transform symbolically from the definition integral for three functions.

Listing 1: Symbolic Laplace Transform via SymPy

```

1 from sympy import *
2
3 t, s, a = symbols('t s a', real=True, positive=True)
4
5 # Helper: integrate e^{-st} * f(t) from 0 to infinity
6 def laplace_from_def(f):
7     return integrate(exp(-s*t) * f, (t, 0, oo))
8
9 # 1. f(t) = 1
10 f1 = S(1)
11 F1 = laplace_from_def(f1)
12 print("L{1} =", F1)                                     # Expected: 1/s

```

```

13
14 # 2. f(t) = e^{at}
15 f2 = exp(a*t)
16 F2 = laplace_from_def(f2)
17 print("L{e^{at}} =", simplify(F2))           # Expected: 1/(s - a)
18
19 # 3. f(t) = t
20 f3 = t
21 F3 = laplace_from_def(f3)
22 print("L{t} =", F3)                         # Expected: 1/s**2
23
24 # Verify numerically at s=5, a=2 for e^{at}
25 F2_val = F2.subs([(s, 5), (a, 2)])
26 print("L{e^{2t}} at s=5 =", F2_val)         # Expected: 1/3

```

Expected output:

```

L{1} = 1/s
L{e^{at}} = 1/(s - a)
L{t} = s**(-2)
L{e^{2t}} at s=5 = 1/3

```

What Did We Learn?

SymPy confirms all three derivations: $1/s$, $1/(s - a)$, and $1/s^2$. The numerical substitution $s = 5$, $a = 2$ gives $1/(5 - 2) = 1/3$ — a quick sanity check you can perform for any formula. SymPy's `integrate` with `oo` correctly handles the improper integral as the limit $T \rightarrow \infty$.

12 Viva-Style Oral Questions

Q1. State the formal definition of the Laplace Transform.

Answer. $\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$, provided the improper integral converges. Here $t \geq 0$ is the time variable and s is a complex parameter (treated as real and positive in most B.Tech applications).

Q2. What does the term “s-domain” (or frequency domain) mean?

Answer. The s -domain is the space in which the transformed function $F(s)$ lives. After applying the Laplace Transform, the independent variable changes from time t (measured in seconds) to the parameter s (measured in 1/second). In electrical engineering, s generalises the concept of frequency, which is why the s -domain is also called the frequency domain.

Q3. Why is the lower limit of the Laplace integral 0, not $-\infty$?

Answer. By convention, the Laplace Transform is designed for *causal* systems — systems that are “switched on” at time $t = 0$ and have no history before that moment. Setting the lower limit to 0 encodes this assumption. Functions defined for all t (including $t < 0$) are handled by the bilateral Laplace Transform, which is a separate (more advanced) topic.

Q4. What is the role of the kernel e^{-st} ?

Answer. The kernel e^{-st} is an exponential damping factor. For $s > 0$ it forces the integrand $e^{-st}f(t)$ to decay to zero as $t \rightarrow \infty$, thereby guaranteeing that the improper integral converges even when $f(t)$ itself grows (e.g. $f(t) = e^{2t}$, provided $s > 2$). Without this kernel, most engineering functions would yield a divergent integral.

Q5. What is the notation convention for time-domain versus s-domain functions?

Answer. The convention is: lowercase letters for time-domain functions ($f(t)$, $g(t)$, $y(t)$) and the corresponding uppercase letters for their Laplace transforms ($F(s)$, $G(s)$, $Y(s)$). The transform operator is always written as $\mathcal{L}$ (calligraphic), never as the plain letter L .

Q6. State and prove the linearity property of the Laplace Transform.

Answer. Statement: $\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)$. **Proof:** Apply the definition, distribute e^{-st} , then use the linearity of the Riemann integral to split into two integrals, each of which equals $aF(s)$ and $bG(s)$ respectively by definition. (See Section 6 for the full proof.)

Q7. Why does $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$ require $s > a$?

Answer. After combining the two exponentials, the integrand becomes $e^{-(s-a)t}$. For this to decay to zero as $t \rightarrow \infty$ we need the exponent $-(s-a)t \rightarrow -\infty$, which requires $s - a > 0$, i.e. $s > a$. If $s \leq a$ the integrand grows or stays constant, making the improper integral diverge.

Q8. What is the improper integral $\int_0^\infty$ as a limit?

Answer. The improper integral is defined as $\int_0^\infty f(t) dt = \lim_{T \rightarrow \infty} \int_0^T f(t) dt$. We first evaluate the ordinary definite integral from 0 to a finite upper limit T , and then take the limit as $T \rightarrow \infty$. The transform exists if and only if this limit is finite.

13 Descriptive Questions

D1. State and explain the formal definition of the Laplace Transform. Define $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st}f(t) dt$. Explain the role of each symbol: $f(t)$ (time-domain function), e^{-st} (damping kernel), the improper integral (limit as upper bound $\rightarrow \infty$), $F(s)$ (s-domain result), and s (complex parameter treated as real in basic applications). Discuss what “convergence” means in this context.

D2. State and prove the linearity property of the Laplace Transform. Write the statement $\mathcal{L}\{af + bg\} = aF + bG$ and prove it step-by-step from the definition integral, citing the linearity of ordinary integration at the key step. Explain the practical significance of linearity for computing transforms of polynomial-exponential combinations.

- D3. Derive $\mathcal{L}\{1\} = 1/s$ from first principles.** Start from $\int_0^\infty e^{-st} \cdot 1 \, dt$, introduce the limit $\lim_{T \rightarrow \infty} \int_0^T e^{-st} \, dt$, evaluate the antiderivative, apply the limit carefully, and state the convergence condition $s > 0$. Show every algebraic step.
- D4. Derive $\mathcal{L}\{e^{at}\} = 1/(s - a)$ and discuss the convergence condition.** Begin from the definition, merge the exponentials, evaluate the integral, and explicitly show why $s > a$ is required. Illustrate with $a = 3$: explain what happens to the integral if $s = 2$ (diverges) versus $s = 5$ (converges to $1/2$).
- D5. Explain the notation conventions for the Laplace Transform and why they matter.** Describe the lowercase/upercase convention, the use of calligraphic $\mathcal{L}$, curly brace notation, and the transform pair symbol $\xleftrightarrow{\mathcal{L}}$. Give at least three examples of how incorrect notation leads to wrong answers in later topics (partial fractions, solving ODEs, inverse transforms).

14 Practice Problems

- P1.** Compute $\mathcal{L}\{3\}$.
Hint: Use $\mathcal{L}\{1\} = 1/s$ and the linearity property. *Answer:* $3/s, \quad s > 0$.
- P2.** Compute $\mathcal{L}\{e^{5t}\}$.
Hint: Apply $\mathcal{L}\{e^{at}\}$ with $a = 5$. State the convergence condition. *Answer:* $1/(s - 5), \quad s > 5$.
- P3.** Compute $\mathcal{L}\{2 + 3e^{-t}\}$.
Hint: Split by linearity; use $a = -1$ in the exponential formula. *Answer:* $\frac{2}{s} + \frac{3}{s+1} = \frac{5s+2}{s(s+1)}, \quad s > 0$.
- P4.** Compute $\mathcal{L}\{e^{-2t} - e^{3t}\}$.
Hint: Apply the formula twice. The binding convergence condition is the more restrictive of the two. *Answer:* $\frac{1}{s+2} - \frac{1}{s-3}, \quad s > 3$.
- P5.** Compute $\mathcal{L}\{t\}$ by integration by parts from first principles.
Hint: Set $u = t$, $dv = e^{-st} \, dt$; evaluate the boundary term at 0 and ∞ carefully.
Answer: $1/s^2, \quad s > 0$.
- P6.** State the correct convergence condition for $\mathcal{L}\{e^{-4t}\}$ and verify by checking the sign of the exponent in the integrand.
Hint: The integrand becomes $e^{-(s+4)t}$; for decay you need $s + 4 > 0$, i.e. $s > -4$. Since we typically take $s > 0$ in practice, the condition $s > 0$ is already sufficient.
Answer: $1/(s + 4), \quad s > -4$.

15 MCQs

MCQ1. The Laplace Transform of $f(t)$ is defined as:

- (a) $\int_0^\infty e^{st} f(t) dt$
- (b) $\int_0^\infty e^{-st} f(t) dt$
- (c) $\int_{-\infty}^\infty e^{-st} f(t) dt$
- (d) $\int_0^1 e^{-st} f(t) dt$

Explanation: The kernel is e^{-st} (negative exponent) and the limits are from 0 to ∞ . Option (a) uses a positive exponent (would diverge); (c) is the bilateral transform; (d) has a finite upper limit.

MCQ2. $\mathcal{L}\{e^{at}\}$ exists for:

- (a) All real s
- (b) $s < a$
- (c) $s > a$
- (d) $s = a$

Explanation: The integrand is $e^{-(s-a)t}$; for convergence we need $s - a > 0$, i.e. $s > a$.

MCQ3. If $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{L}\{g(t)\} = G(s)$, then $\mathcal{L}\{3f(t) - 2g(t)\}$ equals:

- (a) $3F(s) + 2G(s)$
- (b) $3f(s) - 2g(s)$
- (c) $\frac{3F(s)}{2G(s)}$
- (d) $3F(s) - 2G(s)$

Explanation: By the linearity property, $\mathcal{L}\{af + bg\} = aF + bG$ with $a = 3$, $b = -2$.

MCQ4. Which of the following correctly describes the notation convention?

- (a) $\mathcal{L}\{F(t)\} = f(s)$
- (b) $L\{f(t)\} = F(s)$
- (c) $\mathcal{L}\{f(t)\} = F(s)$
- (d) $\mathcal{L}\{f(t)\} = f(s)$

Explanation: Lowercase f in the time domain, uppercase F in the s -domain, and calligraphic $\mathcal{L}$ for the operator.

MCQ5. The kernel e^{-st} in the Laplace integral serves primarily to:

- (a) Amplify the function $f(t)$ at large t
- (b) Convert $f(t)$ from continuous to discrete

- (c) **Ensure convergence of the improper integral for sufficiently large s**
- (d) Differentiate $f(t)$ with respect to t

Explanation: For $s > 0$ the factor $e^{-st} \rightarrow 0$ as $t \rightarrow \infty$, damping the integrand and ensuring a finite result even when $f(t)$ grows.

16 Common Mistakes

Common Mistakes in the Definition of Laplace Transform

Mistake	What Students Do	Correct Approach
Sign error in $\mathcal{L}\{e^{at}\}$	Write $\mathcal{L}\{e^{at}\} = \frac{1}{s+a}$	Merge exponents: $e^{-st}e^{at} = e^{-(s-a)t}$. Result: $\frac{1}{s-a}$, $s > a$
Forgetting the convergence condition	Give $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$ without stating $s > a$	Always state the constraint on s alongside the formula
Writing $F(t)$ or $f(s)$	Use $F(t)$ as the “transformed” function	Use lowercase for time domain ($f(t)$) and uppercase for s -domain ($F(s)$)
Using plain L instead of $\mathcal{L}$	Write $L\{f(t)\}$	Always use the calligraphic $\mathcal{L}\{f(t)\}$
Treating the improper integral as a finite definite integral	Write $[\dots]_0^\infty$ without taking the limit	Define as $\lim_{T \rightarrow \infty} \int_0^T$; evaluate the boundary term via the limit

17 Quick Recap

Quick Recap — Topic 02: Definition of Laplace Transform

- **Formal definition:** $\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$ (improper integral, s sufficiently large).
- **Linearity property:** $\mathcal{L}\{af + bg\} = aF(s) + bG(s)$ — proved directly from the definition integral.
- **First result:** $\mathcal{L}\{1\} = 1/s$, $s > 0$.
- **Exponential transform:** $\mathcal{L}\{e^{at}\} = 1/(s-a)$, $s > a$ (denominator is s minus a).
- **Linear function:** $\mathcal{L}\{t\} = 1/s^2$, $s > 0$ (use integration by parts).
- **Notation conventions:** Lowercase $\rightarrow$ time domain; uppercase $\rightarrow$ s -domain; $\mathcal{L}$ is calligraphic; curly braces always.
- **Convergence conditions:** Always state the lower bound on s alongside the formula — it is part of the answer, not optional.

- **The kernel e^{-st} :** Acts as a damping factor that forces convergence; it is the “calibration” that makes the transform meaningful.